



Remote Access with SSH

Linux has many tools that enable you to access your system remotely; SSH (secure shell) is one. With fully encrypted secure connections, you can remotely copy files, play games, execute commands and do a lot more.

You probably have encountered one of the following situations:

- You want to troubleshoot problems on, or install an application or the LAMP stack on a friend's computer, but can't visit his place.
- You want to run an application on your friend's computer and view it on your own computer. For example, suppose LibreOffice is not installed on your computer, but is on your friend's; so can you use the copy on your friend's computer from your own system?
- You want to copy a file from your friend's computer to your own system, or vice versa.

Linux users will be happy to know you can do these, and a lot more, for free. Let's discuss a very useful remote access tool, SSH (Secure Shell).

The word 'secure' is important here; there are many tools, like telnet, for remote access—but not all are secure. The data transmitted on the network could be intercepted by other users on the same network (called data sniffing). SSH, though, is reliable, convenient, efficient, user-friendly, and most importantly, secured. The SSH client connects to, and facilitates logging in to the remote host (computer). You provide credentials to the remote machine as specified in the server's `sshd_config` file in its `/etc/ssh` directory. When that's verified and accepted by the server, it will execute the command you sent, or log you into a normal command shell on the remote machine. So, the preconditions for using SSH are:

- Both computers must have SSH installed (the server on one and the client on the other). Generally, the SSH client is already installed on Linux computers.

- Both computers must be connected through a network.
- You must have an account (a user ID and password) on the remote host, to log in with.

Basic use

Let us consider a scenario of (only) two computers on a LAN or local network. One, which acts as the server, at the IP address 192.168.1.100, runs CentOS 5.5; the other, my 'local' client computer, is a laptop running Fedora 14, at IP address 192.168.1.101. I have a user account 'binay' on the remote computer. To log in to the remote server, I run `ssh binay@192.168.1.100`. Here, the user name 'binay' is followed by @ and the IP address (or the host-name or fully qualified domain name) of the remote host. The remote server asks for the password; supply the right one, and you are logged in. You could run a simple command like `ls` to list the files in `binay`'s home directory on the remote host.

Now let's move on to copying files between the remote server and the local computer, and vice-versa. Suppose I want to copy the file `riya.odt` from the remote host to my local machine, I just run the following command:

```
scp 192.168.1.100:/home/binay/riya.odt /home/local/
```

To copy a file from the local computer to the remote host, you switch the IP address (of the remote server) to the destination and adjust the target path. For more information, see <http://infhohost.nmt.edu/tcc/help/xfer/scp.html> and http://www.hypexr.org/linux_scp_help.php.

You can also execute commands on the remote computer. For example, let's suppose you want to check the remote